to the board. After uploading the code, open serial monitor to check whether NodeMCU board is connected to Wi-Fi or not.

8. Go back to the created project in Blynk app. Select Joystick from Widget Box and click on Joystick to configure it (Fig. 6). Select V1 pin, and define x and y axis range from 0 to 255 (Fig. 7).

9. Press Play located at the top-right of the screen.

A graphical joystick panel in Blynk app will appear, as shown in Fig. 8. This panel provides digital outputs that keep changing according to the direction you move the joystick to, using your fingers. The joystick has two axes: x and y. Exact direction of the movement is interpreted through the microcontroller (NodeMCU) and the IoT (Blynk).

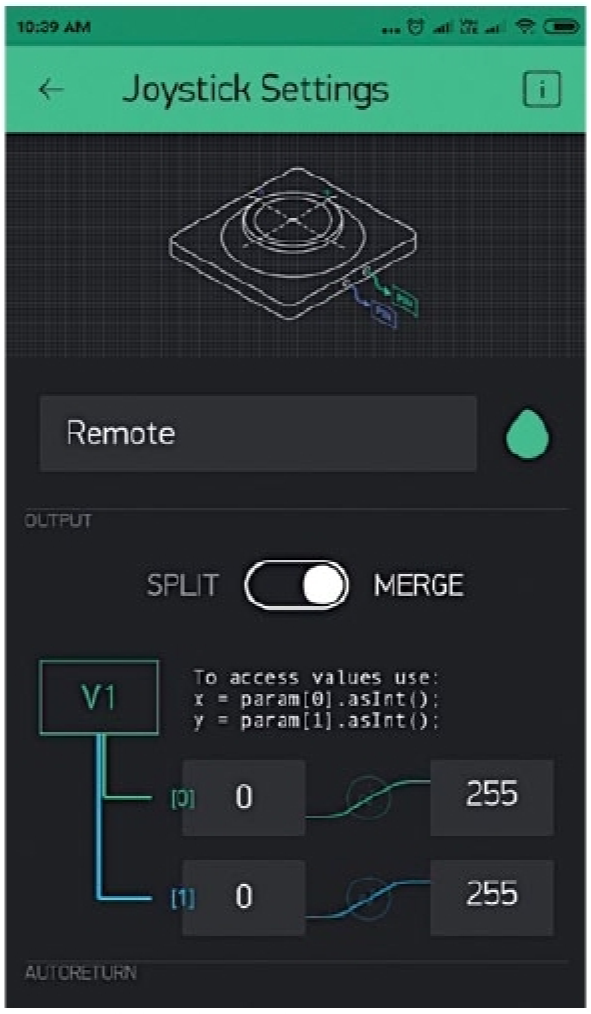

Fig. 7: Joystick settings

Fig. 8: Final widget joystick panel

Software

Circuit operation is done using the software program loaded into the internal memory of NodeMCU. Arduino IDE 1.6.8 is used to compile and upload the program. Software Blynkrobo.ino is written in Arduino language, which allows writing a code within a few lines. The program makes the device communicate with Blynk IoT Application Platform when connected to the Internet through an access point or Wi-Fi router.

To add ESP8266 (NodeMCU) in